MODULE 10 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Multi-VPC, IPAM and IPv6 Adoption

Connecting many VPCs, planning the address space they share, and moving to IPv6 without breaking anything.

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admin@sthithapragna.com

Address space is **shared**. Plan it centrally once, or pay for overlapping CIDRs for years.

Ways to connect VPCs

OPTION	SHAPE	LIMIT THAT DECIDES
VPC peering	Wire between two VPCs	Not transitive; no per-GB processing charge
Transit Gateway	Regional hub with route tables	Per-attachment and per-GB charges
AWS Cloud WAN	Global network from one policy	Built for many Regions and segments
PrivateLink	Doorway to one service	Indifferent to overlapping CIDRs
VPC Lattice	Service network, module 4	Indifferent to overlapping CIDRs

ON THE EXAM

Overlapping CIDRs remove every routing option. Only PrivateLink and Lattice are indifferent to addresses.

VPC peering: limits that decide questions

- **Not transitive.** A peers with B and B with C; A still cannot reach C.
- **No edge-to-edge routing:** no use of a peer's internet gateway, NAT, VPN or endpoints.
- **50 active peerings per VPC** by default, 125 at most. Cross-Region and cross-account work.
- **No per-GB processing charge.** Only normal data transfer.
- **Security group references** work across peering in the same Region.

> ON THE EXAM
>
> Peering is cheaper per byte. Transit Gateway is cheaper per relationship.

Transit Gateway, including what changed

- **Route tables are the segmentation tool.** One gateway can hold many.
- **5,000 attachments,** 50 Gbps burst per VPC attachment, MTU 8500, 1500 over VPN.
- **Security group referencing** across attached VPCs, since late 2024.
- **Network function attachments** connect AWS Network Firewall directly, since June 2025.
- **Encryption support** enforces encryption in transit for all traffic, since November 2025.

> ON THE EXAM
>
> Appliance mode on the inspection attachment keeps both directions of a flow in the same AZ.

Association or propagation?

ASSOCIATION

Which table I read

- Exactly one table per attachment
- Decides where my traffic can go
- Controls what this VPC reaches
- Change it to move a VPC between segments

PROPAGATION

Which tables learn about me

- Can propagate into many tables
- Decides who can reach me
- Controls who sees this VPC
- Or add static routes instead

THE TIEBREAKER

Segments are built from propagation: spokes learn shared services; shared services learn every spoke.

Transit Gateway or Cloud WAN?

TRANSIT GATEWAY

Regional hubs

- One gateway per Region
- Route tables configured per gateway
- Inter-Region links by peering
- Best for one or a few Regions

AWS CLOUD WAN

One global network

- Core network spans many Regions
- Segments defined once in a policy
- Attachments placed into segments by tags
- Connects on-premises sites as well

THE TIEBREAKER

Many Regions with the same segments everywhere, managed from one policy, means Cloud WAN.

Overlapping CIDRs are an organisational failure. IPAM turns address planning into an allocation service.

How VPC IPAM works

- **Pools in a hierarchy:** top-level range, then Regional pools, then pools per environment.
- **Allocation rules** enforce netmask sizes and required tags on each pool.
- **Organizations integration** discovers VPCs across every member account.
- **Pools shared with RAM,** so teams allocate from the plan when creating VPCs.
- **Advanced tier** is needed for pools outside the home Region and allocations to other accounts.

> ON THE EXAM
>
> IPAM prevents overlap at allocation time and reports it afterwards. A spreadsheet does neither.

IPv6 removes address exhaustion and overlap. **Dual-stack** is the bridge; IPv6-only is where it leads.

IPv6 building blocks

PIECE	WHAT IT DOES
Dual-stack subnet	IPv4 and IPv6 together; each subnet gets a /64
IPv6-only subnet	No IPv4 addresses at all; nothing to exhaust
Egress-only internet gateway	IPv6 out to the internet; blocks inbound connections
DNS64 and NAT64	IPv6-only workloads reach IPv4 services via a NAT gateway
Dual-stack load balancers	ALB and NLB serve IPv4 and IPv6 clients
Public IPv4 charge	Every public IPv4 address billed hourly since February 2024

ON THE EXAM

DNS64 and NAT64 are the bridge for IPv6-only workloads: route 64:ff9b::/96 to a NAT gateway.

A dual-stack migration path

01
Add IPv6
Assign an IPv6 block to each VPC, from IPAM.

02
Dual-stack
Give existing subnets and load balancers IPv6.

03
Egress
Egress-only gateway for private IPv6 traffic.

04
IPv6-only
New workloads in IPv6-only subnets, with DNS64 and NAT64.

05
Retire IPv4
Release public IPv4 addresses as clients move.

ON THE EXAM

Dual-stack first, so nothing breaks. IPv6-only for new workloads, where address pressure is worst.

NAT gateway or egress-only internet gateway?

NAT GATEWAY

IPv4 out, and NAT64

- Private IPv4 traffic to the internet
- Translates IPv6 to IPv4 for NAT64
- Hourly charge plus per GB processed
- Needs a public IPv4 address

EGRESS-ONLY GATEWAY

IPv6 out only

- Private IPv6 traffic to the internet
- Stateful: inbound connections blocked
- No hourly or processing charge
- No translation of any kind

THE TIEBREAKER

Private IPv6 traffic to IPv6 destinations should use the egress-only gateway. It costs nothing to process.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Dev must not reach Prod; both reach shared	TGW route table propagation
Two VPCs, huge volume, nothing else	VPC peering on cost
Same segments across many Regions	AWS Cloud WAN
Teams keep creating overlapping CIDRs	VPC IPAM with allocation rules
IPv6-only workload calls an IPv4 API	DNS64 and NAT64
NAT gateway bill, dual-stack subnets	Egress-only gateway for IPv6

ON THE EXAM

Connectivity questions turn on shape and cost. Addressing questions turn on who chooses the range.

Meridian Freight has 40 development VPCs and 25 production VPCs attached to one Transit Gateway, plus a shared services VPC. Today every VPC can reach every other.

Development must not reach production, and both must reach shared services, without adding gateways.

What should the solutions architect do?

A Create a separate Transit Gateway for each environment

B Use separate route tables for development and production, with only shared services propagated into each

C Block development CIDRs in production security groups

D Add network ACLs to every production subnet

◆ PAUSE AND DECIDE

What should the solutions architect do?

A. A gateway per environment

Isolates, but adds gateways, which the scenario rules out.

B. Route tables per segment

No route exists between the segments, so isolation holds at the routing layer.

C. Security group rules

Hundreds of rules to maintain, and a missed one reopens the path.

D. Network ACLs

Same problem per subnet, with stateless rules that are easy to get wrong.

THE SIGNAL

Segmentation on one Transit Gateway is built from route tables and propagation.

> Two Meridian Freight VPCs in the same Region, analytics and the data lake, exchange about 400 TB per month. Nothing else needs to connect to either VPC, and their CIDRs do not overlap.
>
> The company wants the lowest-cost connection.

Which connection should the solutions architect use?

A. VPC peering

B. A Transit Gateway with both VPCs attached

C. AWS Cloud WAN

D. A PrivateLink endpoint service in front of the data lake

◆ PAUSE AND DECIDE

Which connection should the solutions architect use?

A VPC peering

No hourly fee and no per-GB processing; transit limits do not matter here.

B Transit Gateway

Charges per GB processed on all 400 TB, every month.

C Cloud WAN

A global network for two VPCs in one Region, with processing charges too.

D PrivateLink

Charges per GB processed, and exposes one service, not the network.

THE SIGNAL

One pair, huge volume, nothing else. Peering's limits do not apply, and its pricing wins.

> Meridian Freight operates in nine AWS Regions and needs production, development and shared-services segments, isolated the same way everywhere. New VPCs should join the right segment automatically from their tags.
>
> Today, nine Transit Gateways with peering and hand-maintained route tables have drifted apart.

What should the solutions architect recommend?

A. Add an automation pipeline that reconciles the nine Transit Gateway route tables

B. Replace Transit Gateway with a full mesh of VPC peering

C. AWS Cloud WAN with segments defined in the core network policy and tag-based attachment rules

D. A VPC Lattice service network in each Region

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A **Reconciliation pipeline**

Builds by hand what Cloud WAN's central policy already provides.

B **Peering mesh**

Hundreds of connections, no transit, no segments. Far worse.

C **Cloud WAN with policy and tags**

Segments exist once in policy across all Regions; tags place new VPCs.

D **Lattice per Region**

Connects services, not networks, and does not span Regions as a segment.

THE SIGNAL

Many Regions, identical segments, drift, and tag-based joining. Each is a Cloud WAN design goal.

> Meridian Freight's 200 accounts keep producing VPCs with overlapping CIDRs because teams choose their own ranges. Several overlaps have blocked Transit Gateway attachments.
>
> The network team wants ranges allocated centrally per Region and environment, with size limits, and overlaps reported, with the LEAST effort.

What should the solutions architect implement?

A. A shared spreadsheet of ranges, with an AWS Config rule that checks VPC CIDRs

B. An SCP that denies CreateVpc unless the CIDR is in an approved list

C. A Lambda function that deletes any new VPC whose CIDR overlaps

D. VPC IPAM with Organizations integration, Regional and environment pools, allocation rules, shared with RAM

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Spreadsheet and Config**

Detects after the fact; the plan itself still lives outside AWS.

B **SCP with an approved list**

Static list that must be edited for every new range; no allocation, no reporting.

C **Deleting Lambda**

Destructive, custom, and still lets teams choose ranges.

D **IPAM pools and rules**

Teams allocate from the plan; rules enforce size; overlaps are reported.

THE SIGNAL

The cause is teams choosing ranges. Only an allocation service removes the cause.

> Meridian Freight's new EKS cluster runs in IPv6-only subnets to avoid exhausting private IPv4 space. Some pods must call a partner's pricing API, which is reachable only over IPv4 on the internet.
>
> The pods must not have IPv4 addresses.

What should the solutions architect configure?

A. An egress-only internet gateway with a ::/0 route

B. Enable DNS64 on the subnets and route 64:ff9b::/96 to a NAT gateway

C. Assign public IPv4 addresses to the pods' network interfaces

D. A PrivateLink endpoint to the partner's API

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Egress-only gateway

Carries IPv6 only. It cannot reach an IPv4-only destination.

B DNS64 and NAT64

Synthesises IPv6 answers; the NAT gateway translates to the IPv4 API.

C Public IPv4 on pods

Breaks the stated constraint, and adds hourly IPv4 charges.

D PrivateLink

Requires the partner to offer an endpoint service. This API is on the internet.

THE SIGNAL

IPv6-only source, IPv4-only destination. That translation is exactly what DNS64 and NAT64 provide.

> Meridian Freight's private subnets are dual-stack. All outbound internet traffic goes through NAT gateways, which process about 300 TB per month. Most destinations, including the company's main SaaS providers, support IPv6.
>
> The company wants to cut the NAT gateway processing charges with minimal change.

What should the solutions architect do?

A Add an egress-only internet gateway and route ::/0 to it from the private subnets

B Add a NAT gateway in every Availability Zone

C Move the workloads into public subnets

D Replace NAT gateways with NAT instances

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Egress-only gateway for IPv6

IPv6-capable traffic bypasses NAT entirely, with no processing charge.

B More NAT gateways

Can cut cross-AZ charges, but processing per GB is unchanged.

C Public subnets

Exposes workloads, needs public IPv4 addresses, and is a large change.

D NAT instances

Trades a charge for an operational burden and a bottleneck.

THE SIGNAL

Dual-stack subnets plus IPv6-capable destinations: the egress-only gateway is free to process.

What to carry out of this module

- **Peering per byte, Transit Gateway per relationship,** Cloud WAN across many Regions.
- **Segmentation is propagation.** Spokes learn shared services, never each other.
- **IPAM** fixes overlap at the cause: who chooses the range.
- **Dual-stack first,** IPv6-only for new workloads, DNS64 and NAT64 for IPv4 dependencies.
- **Egress-only gateway** moves IPv6 traffic off the NAT gateway meter.

> ON THE EXAM
>
> Public IPv4 addresses have been charged hourly since February 2024. That cost now drives IPv6 decisions.